AI and Blockchain-Based System for Real-Time Apple Quality Grading and Traceability

Varsha Dange
Vishwakarma Institute of Technology
Pune, India
varsha.dange1@vit.edu

Aditi Patil
Vishwakarma Institute of Technology
Pune, India
aditi.patil241@vit.edu

Abhishek Patil
Vishwakarma Institute of Technology
Pune, India
abhishek.patil241@vit.edu

Aditya Kachwaha
Vishwakarma Institute of Technology
Pune, India
aditya.kachwaha24@vit.edu

Abhinandan Bardia
Vishwakarma Institute of Technology
Pune, India
abhinandan.bardia24@vit.edu

Ved Aghav
Vishwakarma Institute of Technology
Pune, India
ved.aghav24@vit.edu

***Abstract*—A blockchain-secured mobile application integrates artificial intelligence to automate the quality grading of apples. Transfer learning is used to develop two deep learning models, MobileNetV3 and EfficientNet-B0, for classifying apples into predefined grades (A, B, and C) based on color, texture, and surface defects. The performance of both models is evaluated to determine the architecture that provides higher accuracy and better generalization in real-world scenarios.**

The application enables farmers to capture or upload apple images and obtain instant grading results. The predicted grade and associated metadata, including farmer details, timestamp, and location, are securely recorded on a decentralized blockchain ledger, ensuring transparency, traceability, and immutability. Each batch is assigned a unique QR code that allows consumers to verify product origin and provide feedback.

This approach enhances trust among stakeholders, reduces fraud, improves quality assurance, and increases efficiency in fruit distribution networks. Additionally, the platform provides farmers with access to nearby warehouses and collection centers, enabling faster post-harvest decision-making. Model performance is further evaluated using precision, recall, and F1-score to ensure balanced classification across all grades.

***Index Terms*—Fruit Quality Grading, MobileNetV3, EfficientNet-B0, Transfer Learning, Deep Learning, Image-Based Classification, Color and Texture Analysis, Surface Defect Detection, Real-Time Grading, Blockchain Ledger, Decentralized Traceability, QR Code Generation, Farmer Metadata, Location Tracking, Supply Chain Transparency, Warehouse Locator, Mobile Application**

I. INTRODUCTION

Grading is a critical activity in the quality assessment of apples, as it determines market value, consumer acceptance, and storage suitability. In orchard environments and physical markets, grading is typically performed based on visible attributes such as color, texture, firmness, and surface defects. However, this approach is often inconsistent due to human subjectivity, varying lighting conditions, and non-uniform grading standards. As a result, misclassification and pricing disputes frequently occur, leading to reduced trust among stakeholders.

This work addresses these limitations by proposing an integrated solution that combines image-based classification with blockchain-enabled traceability through a mobile application. Two deep learning models, MobileNetV3 and EfficientNet-B0, are fine-tuned using transfer learning to classify apples into grades A, B, and C based on visual characteristics.

After grading, metadata such as farmer identity, timestamp, and GPS location is securely recorded on a decentralized ledger, ensuring tamper-proof storage. A unique QR code is generated for each batch, allowing verification of product origin and quality. Additionally, a warehouse locator feature assists farmers in identifying nearby cold storage facilities, enabling efficient post-harvest handling.

A. Research Gap

Traditional grading approaches rely heavily on single-model architectures, controlled laboratory datasets, or centralized databases. These methods limit real-world applicability and lack traceability. Moreover, limited research focuses specifically on apple grading using lightweight yet accurate architectures such as MobileNetV3 and EfficientNet-B0. Existing solutions also provide minimal support for real-time decision-making related to storage and logistics. These gaps highlight the need for an integrated approach that ensures accurate grading, secure traceability, and actionable insights for post-harvest management.

B. Objectives

1) Develop a reliable apple grading framework that improves accuracy and transparency while enhancing supply chain efficiency.
2) Enable real-time grading through AI-based image analysis to provide consistent and unbiased results.
3) Ensure secure storage of grading metadata using blockchain for immutability and traceability.
4) Generate QR-based identifiers to allow stakeholders to verify product origin and handling details.
5) Provide location-based warehouse recommendations to optimize logistics and reduce post-harvest losses.
6) Minimize mislabeling and improve quality assurance for consumers and distributors.

7) Integrate AI-based grading with blockchain-backed traceability to strengthen trust across the supply chain.

II. LITERATURE REVIEW

A. Traditional and Early Fruit Grading Methods

Initially, fruit quality grading in an automated capacity stemmed from classical machine learning, image processing, preprocessing, segmentation and handcrafted feature extraction. For instance, Peng et al. [1] demonstrate that image processing can help grade apples and therefore serve as a future means to grade. These findings achieved relatively accurate results in a lab setting, but since these methods are based on manual feature extraction, they are not stable and generalized in the real world with varied lighting and backgrounds. These learning-based techniques, instead, were applied to certain fruits down the line, but they were not useful to scale for larger, more heterogeneous data sets [11].

B. CNN-Based Fruit Grading and Comparative Model Studies

The deep learning revolution allowed for an even deeper understanding of automatic fruit grading systems via end-to-end feature learning. Convolutional neural networks (CNNs) applied to fruit grading systems exist for apples [2], pomegranates [3], citrus [4] and mangoes [6], often with higher accuracy than non-CNN/deep learning systems. Note that Patel and Patil [2] acknowledge the power of fine-tuning certain CNN architectures to achieve high accuracy for grading purposes of comparison post-harvest.

A common way to render highly accurate comparisons among different kinds of CNNs is via transfer learning. For example, Narvekar and Rao [3] acknowledge the increased accuracy and reduced epochs to convergence via backbones of CNNs pretrained on ImageNet versus building networks from the ground up. In addition, transfer learning illustrates that more extensive network architectures with more layers—i.e., DenseNet—can detect subtle texture variances and surface damage better than models, which increases the accuracy [6]. Furthermore, studies on multi-fruit grading acknowledge the transferability of CNNs when tested across different kinds [13], [15].

C. Lightweight Models, Explainability, and Research Gap

Aside from accuracy, computational effectiveness is another major consideration of comparative studies. In this sense, lightweight CNN architectures (MobileNet-based, specifically) constitute a good trade-off between performance and efficiency, which is favored for real-time, resource-limited agricultural settings [7].

A few recent developments in the field include explainable AI for more effective model comparisons. Explainable CNN architectures enable the researcher to better understand which predictions stem from which functions of each layer, yielding insights into performance differential and reliability of models [5]. Furthermore, surveys and benchmarking studies establish that CNNs based on transfer learning are the most popular models in comparative studies as of yet for automated grading solutions [10], [14].

However, many studies in the field compare a handful of models at best or studies focus on one model for one fruit type. Therefore, there exists no generalized experimental approach to comparing deep learning architectures beyond several at one time and in a systematic way. This justifies the need for this work.

III. METHODOLOGY/EXPERIMENTAL

A. Dataset Collection and Preprocessing

The real apple image dataset was created from a combination of images taken in orchards, cold-storage and market facilities, and public domain libraries. Images were taken of different varieties of apples, different rates of maturation, superficial bruising, natural blemishes, cupping, and changes in lighting. All images were resized to input dimensions accepted by MobileNetV3 and EfficientNet-B0 networks and normalized to maintain color consistency across the datasets. For better stability and generalization, data augmentation was applied in the form of rotation, brightness and contrast adjustment, zoom, noise addition, and random cropping.

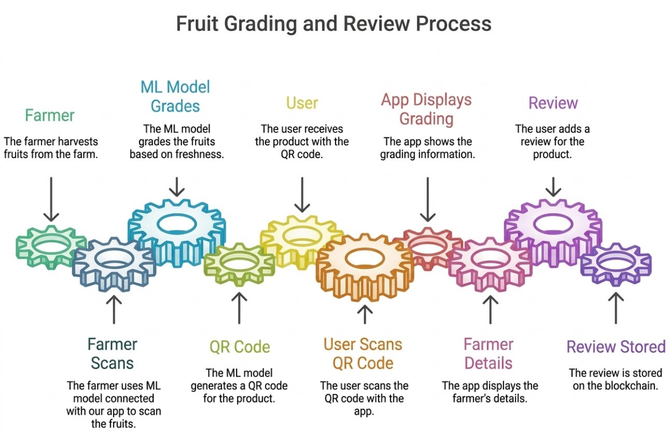

Fig. 1. Workflow of the fruit grading, QR traceability, and blockchain-based review system.

B. Model Training and Comparison

Transfer learning was applied through MobileNetV3 and EfficientNet-B0 with pre-trained ImageNet weights. The classification layers were modified to accommodate three levels of classification quality assessment: Grade A, Grade B, and Grade C. Thereafter, multiple layers were re-trained to fit the apple-specific aesthetics (color consistency, surface smoothness, bruising). The training was conducted using learning rate scheduling, dropout, batch normalization, and early stopping. For precise evaluation of the performance of the models, precision, recall, and F1-score for each class in the dataset were also calculated. This helped to maintain unbiased performance of the model across all grade classes.

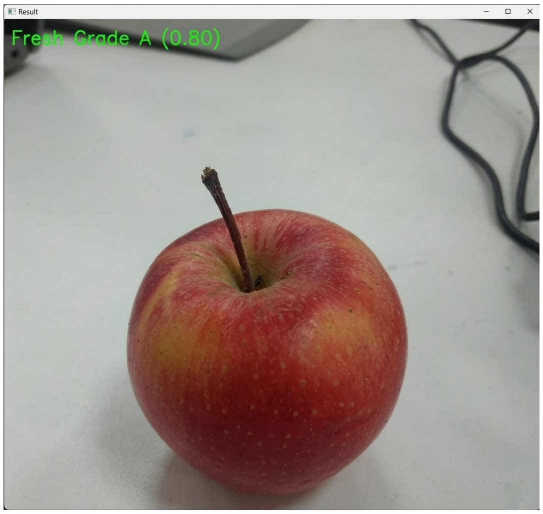

Fig. 2. Model output demonstrating classification of the apple as Grade A with a confidence level of 0.80.

C. Real-Time Grading Workflow

The application of the suggested model translates to a mobile solution for on-field apple grading. The farmer captures or uploads an apple image through the app, which preprocesses and performs inference in the background. The application presents the three grade probabilities while predicting quality within a few seconds so that farmers can receive fast, impartial grading without reliance on external parties in the field.

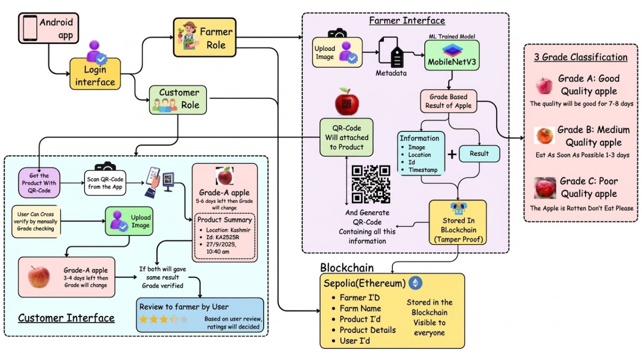

Fig. 3. System Architecture of the Proposed Apple Quality Grading and Traceability System.

D. Metadata Capture and Blockchain Storage

All grading-related data is securely stored to ensure transparency and to prevent retroactive manipulation after grading. All elements of the grading process are saved as metadata (who graded it, when it was graded, where it was graded—GPS coordinates). This information is then structured and transmitted securely to a blockchain-enabled decentralized ledger along with machine learning outputs and confidence levels for the generated grade.

E. QR-Based Traceability and Post-Harvest Support

Each graded lot is assigned a unique QR code that links it to the blockchain transaction. Scanning the QR code allows stakeholders in the supply chain to validate the grading and origin of the apples. Additionally, a warehouse locator module integrates with the system's GPS to identify nearby cold-storage warehouses and collection centers, enabling improved post-harvest decisions and reduced wastage.

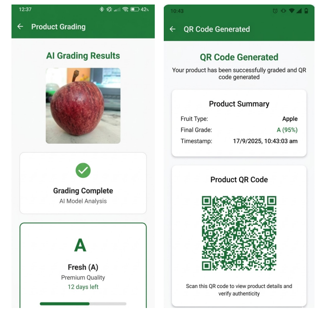

Fig. 4. AI grading results display with fruit quality analysis (left) and QR code generation interface (right).

F. System Integration

All modules are integrated into a unified mobile system. Lightweight AI processing occurs on-device, while blockchain processing is handled through cloud/edge services. This hybrid architecture enables low-latency, scalable, and reliable operation in real-world agricultural environments.

IV. COMPARATIVE ANALYSIS TABLE

System	Objective	Strengths	Limitations	Contribution
Peng et al.	Classify grades	Low cost	No feature learning	DL + blockchain
Patel and Patil	Detect diseases	High accuracy	No grading	End-to-end grading
Narvekar and Rao	Multi-fruit grading	Strong accuracy	GPU dependent	Lightweight + blockchain
Kona et al.	Defect detection	Mobile-friendly	Weak verification	QR + blockchain
Hybrid Model	Predict maturity	High accuracy	No full pipeline	Complete grading system
DenseNet	High accuracy	Strong learning	Heavy computation	Edge + audit trail
ESP32	Low-cost sorting	Fast hardware	Low resolution	Cloud + blockchain
Blockchain-only	Traceability	Secure logs	No AI grading	Unified pipeline

Table 1: Comparative Study of Existing Fruit Grading Methods and the Proposed Integrated AI–Blockchain Solution

The standard deviation and variance of the model's performance were also analyzed.

V. RESULTS AND DISCUSSIONS

Experiments evaluate the performance of the proposed approach in terms of model accuracy, blockchain traceability, QR-based verification, and warehouse recommendations. The results demonstrate that the integrated framework enhances reliability and transparency in grading and post-harvest decision-making.

Three deep learning models, MobileNetV3, EfficientNet-B0-B0, and ResNet50, are trained on the apple dataset. MobileNetV3 achieves the highest accuracy of 93%, compared to 90.8% for EfficientNet-B0-B0 and 89.2% for ResNet50. It also requires fewer training samples and provides the fastest inference time, making it highly suitable for real-time mobile deployment. EfficientNet-B0-B0 performs well in detecting fine defects but exhibits slower inference. ResNet50, due to its heavier architecture, is less suitable for mobile applications. Overall, MobileNetV3 provides the best balance between accuracy, efficiency, and deployment feasibility.

Factor	**MobileNetV3**	**EfficientNet-B0-B0**	**ResNet50**
Accuracy	≈93%	≈90.8%	≈89.2%
Images Required	∼2000	∼3000–3500	∼4000+
Model Size	5–6 MB	16–20 MB	95–100 MB
Training Time	Fastest	Moderate	Slowest
Inference Speed	Very Fast	Fast	Slow
Computational Cost	Very Low	Low–Moderate	High
Suitability	Excellent	Good	Poor
Generalization	Very Strong	Strong	Moderate
Fine Defect Detection	Strong	Strongest	Moderate
Best Use Case	Real-time mobile	Cloud grading	GPU research

Table 2: Performance Comparison of Deep Learning Models Used in the Proposed Apple Grading System

In addition to accuracy, precision, recall, and F1-score were used to evaluate the models. In this context, MobileNetV3 recorded a precision of 0.92, a recall of 0.91, and an F1-score of 0.915, which indicates a balanced classification across all grades. From the confusion matrix results, it is observed that Grade A apples are classified with the highest accuracy, while some overlap occurs between Grade B and Grade C due to similarities in surface characteristics. In tests conducted on actual orchard floors, farmers captured images of apples and received grading results within seconds, thereby demonstrating the usability of the system under variable lighting and field conditions. The blockchain module generated grading records containing farmer identity, timestamp, grade, and GPS location as immutable entries, ensuring traceability without tampering. QR codes enabled buyers and consumers to instantly verify provenance. Additionally, the warehouse locator proved effective in identifying nearby cold-storage and collection centers, reducing spoilage and improving post-harvest handling.

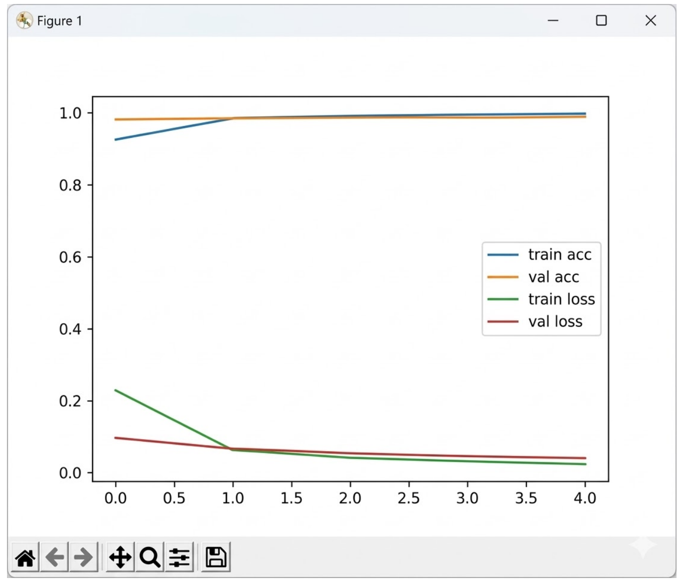

Fig. 5. Training and validation accuracy and loss curves of the fruit grading model.

Overall results confirm that MobileNetV3 is the most efficient and accurate model among those evaluated. The proposed system significantly improves apple traceability, quality control, and distribution efficiency. Statistical evaluation further confirms the consistency and reliability of MobileNetV3 for real-time agricultural applications.

VI. FUTURE SCOPE

The proposed approach can be extended to incorporate IoT technologies for monitoring the cold chain through temperature and humidity sensors. This enables real-time tracking during storage and transportation, improving product quality and reducing losses.

VII. CONCLUSION

This paper presents an automated apple grading system that integrates image-based classification with blockchain-based traceability and mobile decision support to enhance trust in the apple supply chain. Among the evaluated models, MobileNetV3 achieved the highest accuracy while maintaining efficiency suitable for real-time applications.

The system records grading results along with metadata such as farmer details, timestamps, and GPS locations on a blockchain ledger, ensuring transparency and tamper-proof recordkeeping. Each batch of apples is associated with a unique QR code, allowing consumers to verify authenticity and origin instantly. Additionally, the warehouse locator feature assists farmers in identifying nearby storage facilities, reducing spoilage and improving post-harvest management.

Overall, the system enhances grading consistency, strengthens trust among stakeholders, and improves operational efficiency across the supply chain. The inclusion of statistical evaluation further supports the reliability of the proposed solution.

VIII. ACKNOWLEDGMENT

The authors would like to thank Vishwakarma Institute of Technology, Pune, for providing the necessary resources and support. The authors also express gratitude to the faculty members of the Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning) for their valuable guidance. Appreciation is extended to all contributors who provided feedback and participated in discussions that helped refine this work.

IX. REFERENCES

REFERENCES

[1] J. Peng, J. Li, X. Li, L. Fang, P. Zhang, and Y. Lv, "Research on Image Recognition and Grading Method of Apple Based on Machine Vision," in 2022 2nd International Conference on Computer Science, Electronic Information Engineering and Intelligent Control Technology (CEI), Nanjing, China, 2022, pp. 511–515.

[2] H. B. Patel and N. J. Patil, "Enhanced CNN for Fruit Disease Detection and Grading Classification Using SSDAE-SVM for Postharvest Fruits," IEEE Sensors Journal, vol. 24, no. 5, pp. 6719–6732, 2024.

[3] C. Narvekar and M. Rao, "Fruit and Vegetable Grading with Transfer Learning and Convolutional Neural Networks for Better Productivity," in 2023 International Conference on Advanced Computing Technologies and Applications (ICACTA), Mumbai, India, 2023, pp. 1–5.

[4] R. M. Math, N. V. Dharwadkar, V. Sajjanar, and M. N. Deshmukh, "Deep Learning for Precision Grade-Quality Classification of Pomegranates," in 2024 International Conference on Innovation and Novelty in Engineering and Technology (INNOVA), Vijayapura, India, 2024, pp. 1–6.

[5] S. Dutta, S. G. Neogi, D. Chanda, A. Pramanik, Ö. Girgin, and E. L. Öncül, "FruitQ-GradeX: Determining Fruit Quality and Grading with Explainable Deep Learning," in 2025 12th International Conference on Reliability, Infocom Technologies and Optimization (ICRITO), Noida NCR, India, 2025, pp. 1–7.

[6] M. S. Morshed, S. Ahmed, T. Ahmed, M. U. Islam, and A. B. M. Ashikur Rahman, "Fruit Quality Assessment with Densely Connected Convolutional Neural Network," in 2022 12th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, 2022, pp. 1–4.

[7] M. S. R. Kona, A. Guvvala, V. V. L. Eedara, M. S. Gowri, and V. Aluri, "Mango Fruit Defect Detection Using MobileNetV2," in 2024 International Conference on Emerging Innovations and Advanced Computing (INNOCOMP), Sonipat, India, 2024, pp. 22–27.

[8] V. Agrawal, H. Barhate, H. Waghmare, K. Bhattad, and S. D. Shelke, "Fruit (Orange) Quality Inspection using Fully Convolutional Network," in 2024 MIT Art, Design and Technology School of Computing International Conference (MITADTSoCiCon), Pune, India, 2024, pp. 1–6.

[9] C.-L. Chen and C.-Y. Chang, "Implementation of Apple Quality Automatic Classification System Based on Deep Learning Technique," in 2025 IEEE International Conference on Consumer Electronics – Taiwan (ICCE-Taiwan), Kaohsiung, Taiwan, 2025, pp. 627–628.

[10] P. Singh, A. Rana, T. Choudhury, and N. Tripathi, "AI & ML-Based Smart Non-Disruptive Methods for Post-Harvest Fruit Grading: State-of-the-Art," in 2024 MIT Art, Design and Technology School of Computing International Conference (MITADTSoCiCon), Pune, India, 2024, pp. 1–6.

[11] K. Ahmed, Kajal, M. K. Dubey, and D. K. Pandey, "Machine Learning Enabled System Architecture for Automatic Grading and Sorting of Walnut Fruits: A Review," in 2024 International Conference on Electrical Electronics and Computing Technologies (ICEECT), Greater Noida, India, 2024, pp. 1–6.

[12] N. Kundu, Sonam, G. Rani, and V. S. Dhaka, "A Deep Learning Based System for Automatic Sorting and Quality Grading of Citrus Fruits," in 2024 International Conference on Emerging Trends in Networks and Computer Communications (ETNCC), Windhoek, Namibia, 2024, pp. 1–5.

[13] N. E and V. G, "Convolutional Neural Network-Based Multi-Fruit Classification and Quality Grading with a Gradio Interface," in 2024 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES), Chennai, India, 2024, pp. 1–7.

[14] K. Dakwala, V. Shelke, P. Bhagwat, and A. M. Bagade, "Evaluating Performances of Various CNN Architectures for Multi-Class Classification of Rotten Fruits," in 2022 Sardar Patel International Conference on Industry 4.0, Mumbai, India, 2022, pp. 1–4.

[15] L. Yi and J. Zhong, "Apple Grade Classification Based on Convolutional Neural Network and Machine Learning," in 2024 International Conference on Neuromorphic Computing (ICNC), Chongqing, China, 2024, pp. 1–5.